# Building a Digital Operating and Traceability Platform

## for the Iloilo Rice Processing Complex

Iloilo Startup Hackathon Challenge

October 14, 2026 • Office of the Provincial Agriculturist



# The system is constrained at both ends

## PEAK

Congested slots
Delayed procurement

## OFF-SEASON

Idle capacity
High fixed costs

**24–28%**

of income drained by electricity





**Paper workflows obscure prices, queues, utilization and the farmer-to-buyer trail.**

# The plant behind the platform

**5**

Mechanical dryers

**50 t**

Milling capacity per day

**1,100 t**

Storage capacity

---

**Pilot scale: about 350–400 hectares of cooperative farms in Pototan**

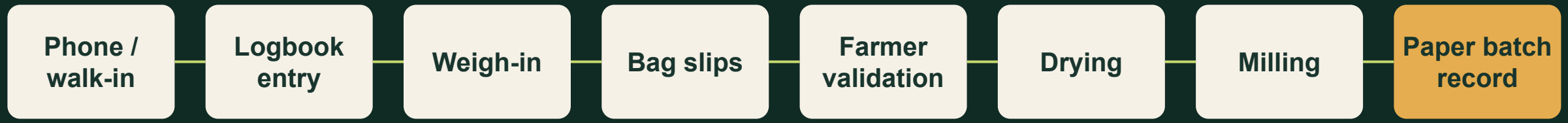
**Losses in 2024 and 2025. First profitable quarter only in early 2026.**
**Private mills still win on speed and price.**

# One platform. One rice value chain.

- Smallholder farmers
- Cooperatives & irrigators
- Retailers, wholesalers & institutions
- Indigent households — ₱20/kg program
- IRPC plant personnel
- Municipal Agriculture Offices



# Today, every handoff creates friction




No advance slots or prices • Manual consolidation • Delayed cost visibility • No live queue

**The result: decisions arrive after the operational moment has passed.**

# Rules the platform must respect

| | |
|---|---|
| **PRICES** | Posted on the office board and weekly on Facebook |
| **PRICE BASIS** | Market price now; NFA price once IRPC is an LEE |
| **VALIDATION** | Farmer confirms at weigh-in and again after drying |
| **COST VIEW** | Energy and margin seen only after month-end books |
| **RECEIPTS** | Audit-ready records that never replace official receipts |

# “A two-sided digital platform that streamlines operations, improves transparency, and maximizes IRPC’s utilization year-round.”

Fill idle capacity

Speed procurement

**Trace every batch**

Lower cost per kilo

Sustain subsidy surpluses

# Five modules build the operating spine

### 01 Booking & Queue

Bookings, alerts, live dashboard

### 02 Real-Time Costing

Energy and margin by batch

### 03 Traceability

Farmer-to-buyer batch history

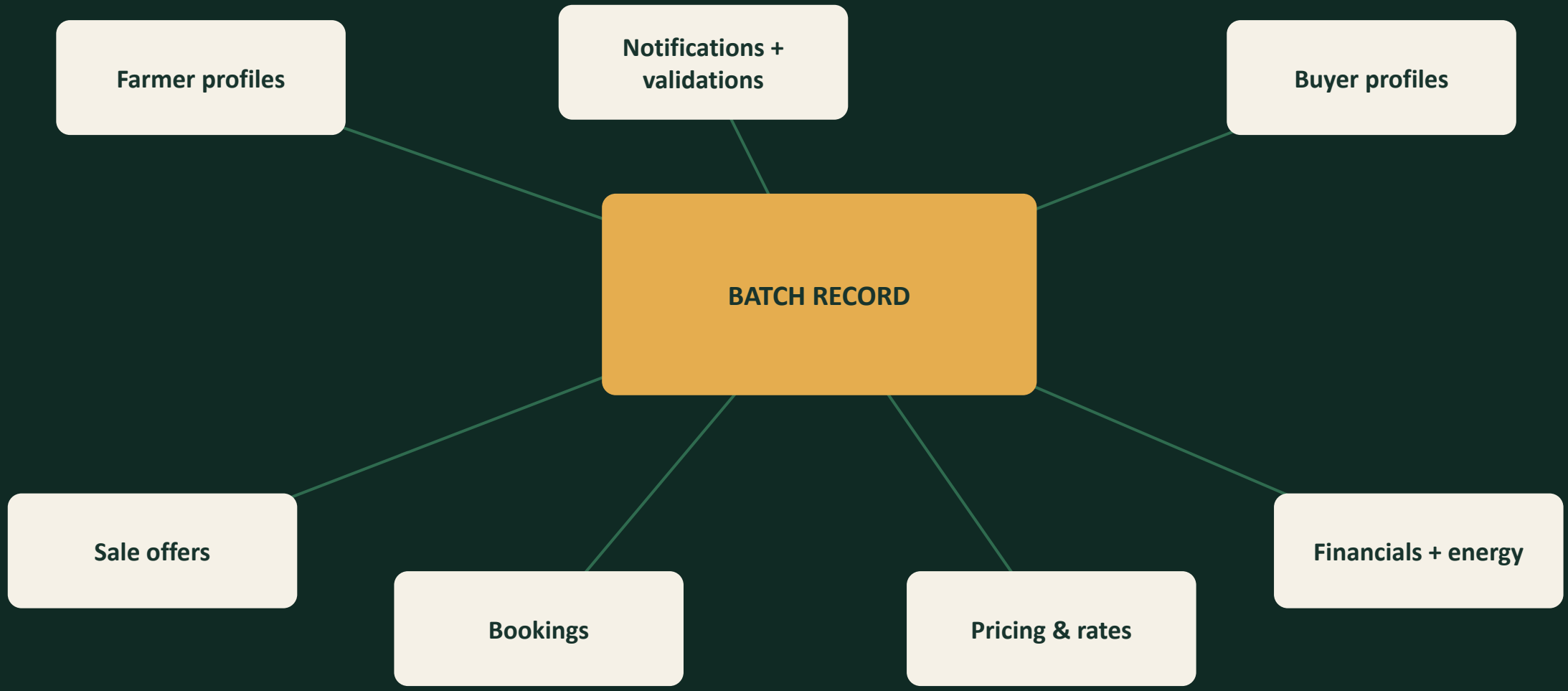
### 04 Pricing & Rates

Daily palay and service fees

### 05 Harvest Forecasting

Plan capacity; attract off-peak volume

# A batch is the digital thread



# Trace every batch to its first buyer

A first-hand buyer is anyone who buys directly from, or takes first delivery from, the IRPC rice mill.

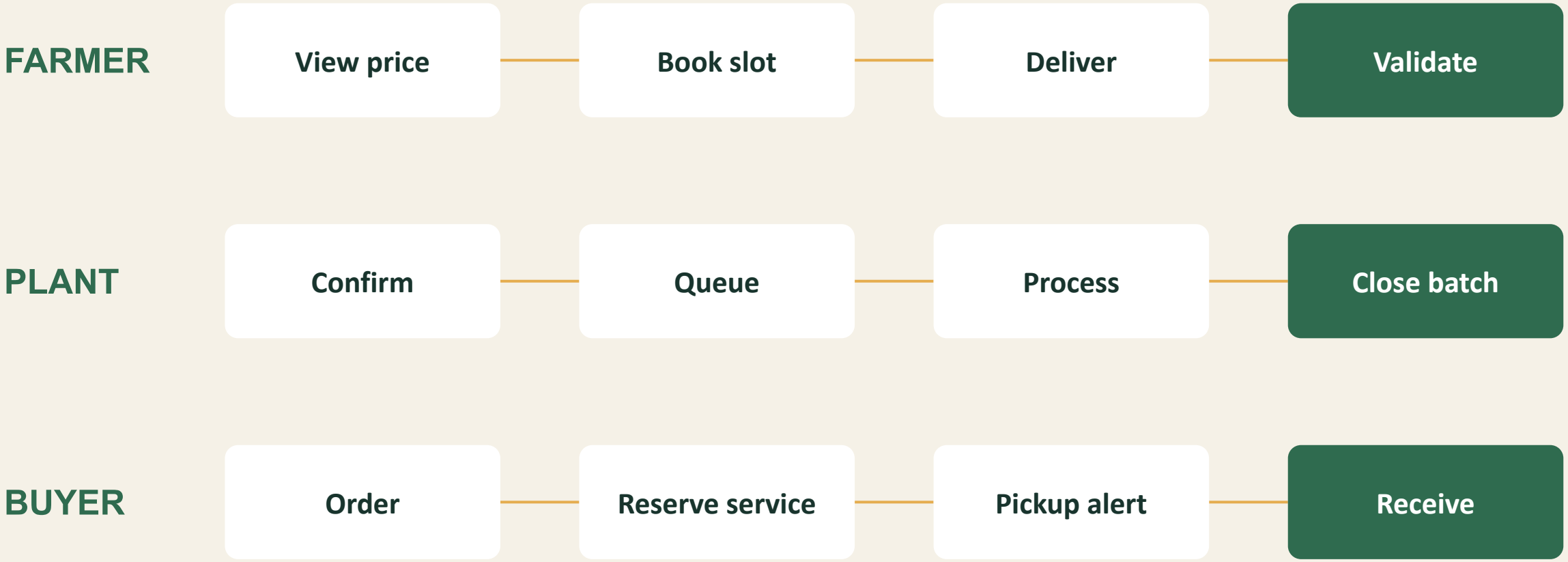
- **Wholesalers**
- **Stores**
- **Municipal drop-points**
- **Walk-in buyers**
- **₱20/kg program**

**Farmer → drying → milling → first-hand buyer**

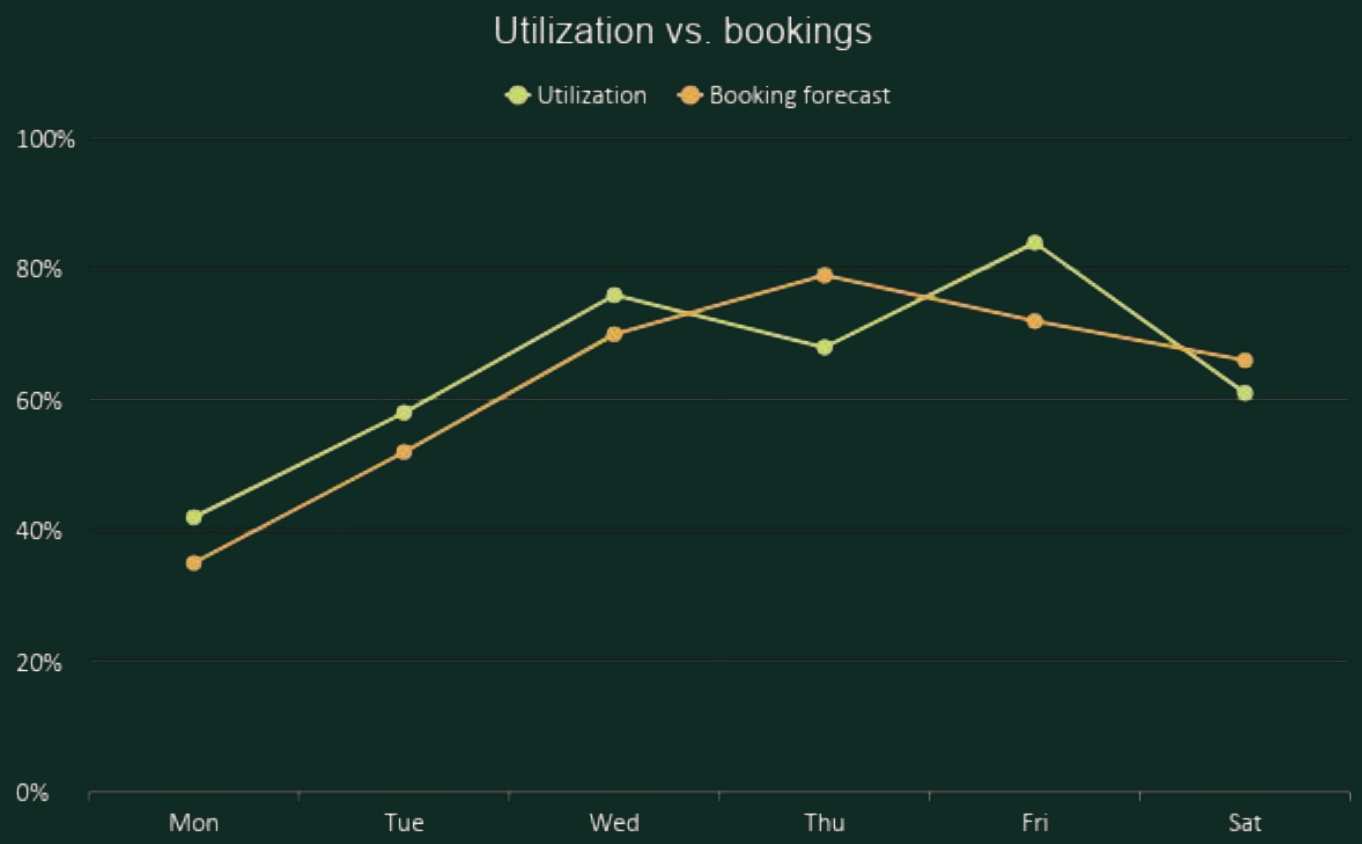
# Access follows responsibility

| | |
|---|---|
| **FARMERS** | Prices • offers • slots • delivery validation |
| **BUYERS** | Prices • service booking • pickup alerts |
| **PLANT PERSONNEL** | Queues • equipment • batch records |
| **MAO STAFF** | Reports • subsidies • validation |
| **ADMINS** | Configuration • users • audit logs |

# Three flows meet at one plant queue



# See the plant as it operates




**LIVE QUEUE**
**12 batches**

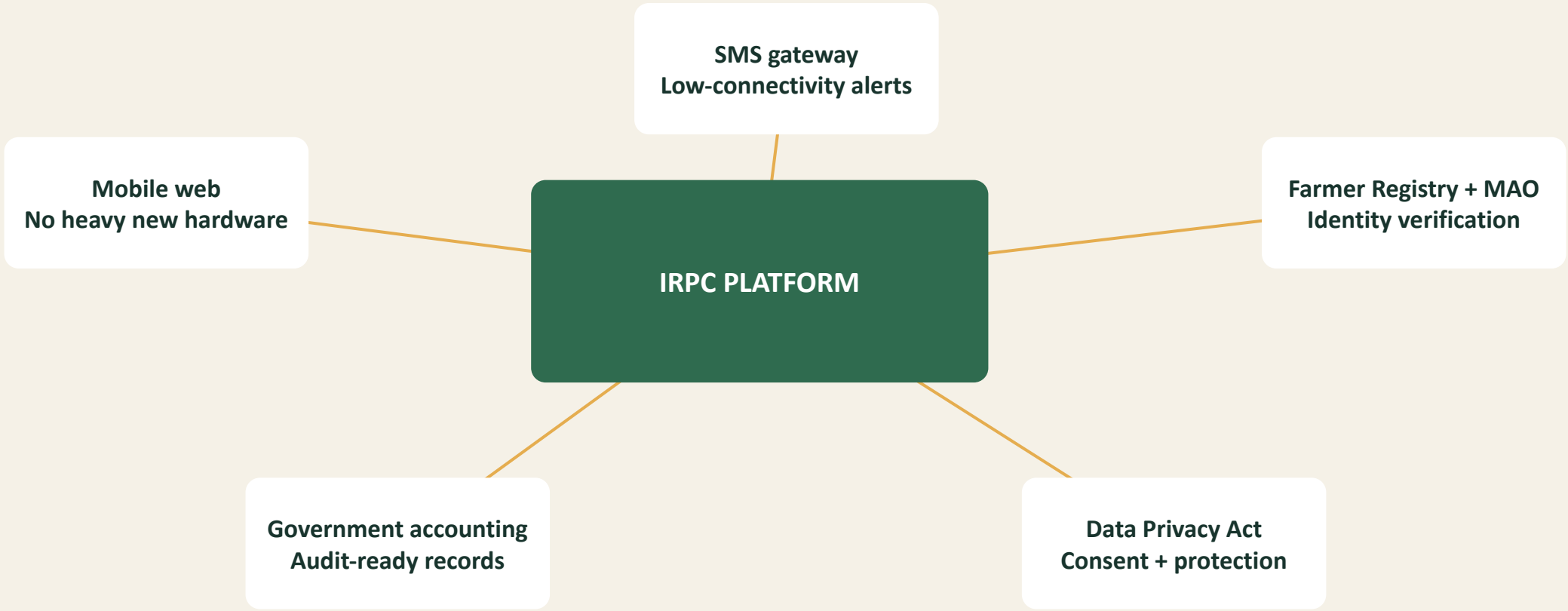
**ENERGY / BATCH**
**₱18,420**

**GROSS MARGIN**
**₱31,580**

**CASH + SUBSIDY**
**Updated today**

Figures shown are illustrative

# Lightweight by design. Accountable by default.



# Pilot first. Integrate with confidence.

**PHASE 1 — PILOT**

Booking + queue
Pricing display

**PHASE 2**

Batch costing
Notifications

**PHASE 3**

Traceability
Harvest forecasting

**PHASE 4**

OPAg MoRProGRes integration

**POTOTAN • next harvest season • farmer cooperative pilot**



# Prove it in Pototan. Then scale province-wide.

1 Pilot with cooperatives next harvest

2 Learn through structured user feedback

3 Refine, scale, and support IRPC’s transition to a Local Economic Enterprise

# Design the mitigations into the product

| | |
|---|---|
| **Connectivity** | → SMS fallback + simple UI |
| **Adoption** | → Training + Hiligaynon / English |
| **Data accuracy** | → Validation + immutable logs |
| **Cost overrun** | → Phased delivery + lean stack |
| **Integration** | → Early APIs + external testing |

# Data teams can work with

1 Monthly financial statements, Aug 2024 – Mar 2026

2 Feasibility study with private-mill benchmark

3 Business plan and 2027–2031 projections

4 Palay procurement by municipality, 2021–2023

5 20 regular buyers with annual volumes

6 Plant specs, process flow and service rates

7 PSA provincial and regional rice production data

Personal data shared in anonymized form

# What we need programmers to decide

1 Tech stack + SMS gateway?

2 Low-data UI + offline approach?

3 Traceability + costing data model?

4 Security + privacy controls?

5 Pilot monitoring tools?

6 Scale + maintenance plan?

# Let’s build a platform that empowers Iloilo’s rice value chain.

**NEXT**
**Align MVP scope • Finalize tech approach • Prepare pilot environment**

Aileen Mae S. Tanalgo

Assistant Plant Manager, OPAg IRPC

## Thank you

